
SUMMARY

Software Engineer with a strong foundation in full-stack development, distributed systems, and infrastructure automation. Proficient in **Java, Python, and C**, with experience building scalable applications using **gRPC, ZeroMQ, MongoDB**, and **DataFusion**. Skilled in **Agile** methodologies, **CI/CD pipelines**, and containerization with **Docker**. Passionate about solving real-world problems in collaborative, **fast-paced** environments and learning modern software engineering practices.

PROFESSIONAL EXPERIENCE

Siemens | *Software Engineer Intern*

Pasadena, CA | June 2024 - September 2024

- Collaborated with senior engineers to design a monitoring and remediation tool for **MySQL** and other apps on **AWS EC2**.
- Automated system health checks and monitoring pipelines using **Rust** and **Go**, reducing toil by **33%**.
- Deployed proactive alerting systems and optimized logging tools by leveraging telemetry data to cut incident response time.
- Built a lightweight **pub/sub** system and standardized health metrics for seamless cross-service monitoring.
- Improved system efficiency and resilience by **15%** through performance optimizations and proactive issue resolution.

theCoderSchool | *Coding Mentor*

Pasadena, CA | August 2020 - Present

- Taught **Python, Java**, and **C** through project-based lessons tailored to individual skill levels.
- Guided students in **debugging**, algorithm development, and coding best practices

PROJECTS

PantryPal | *Java, JavaFX, MongoDB, Docker*

December 2023 | [Github](#)

- In a team of 6, built a full-stack application using **Java, MongoDB**, and OpenAI **APIs** (GPT, DALL·E, Whisper).
- Built modular **RESTful APIs** for recipe generation, voice input, and storage, ensuring **scalability** and **maintainability**.
- Applied **Agile** methodology and **CI/CD** pipelines with **GitHub Actions**, enabling **fast iteration** and **bug-free** releases.
- Containerized the application with **Docker** to have consistent development and deployment across environments.
- Followed **MVC** and **Design by Contract** principles to ensure modularity, reducing code complexity by **20%**.

SolarCross | *Python, OpenCV, C++, HTTP*

December 2024 | [Github](#)

- Built a modular **AI** pipeline in **Python** with **OpenCV**, designed for real-time performance and easy extension.
- Optimized video stream processing using **multithreading**, with scalable logging and error handling.
- Processed **MJPEG HTTP** streams in real-time to enable live feedback in a solar tracking system.

LectureSlidesCollaborateChatroom | *React, Node.js*

July 2025 | [Github](#)

- Built a real-time **React** chatroom with per-slide and independent slide navigation using **Socket.IO**.
- Developed **Node.js** backend to handle PDF/PPT uploads and enabling synchronized commenting.

TECHNICAL SKILLS

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| • Core Programming Languages: Java, Python, C/C++ | • Tools: GitHub, MATLAB, VS Code, Docker, Cmake |
| • Familiar with: HTML, CSS, JavaScript, Assembly, Verilog, SystemVerilog, RobotC, Pascal, Haskell, Go, Rust | • Databases: MySQL, MongoDB (NoSQL) |
| | • Libraries/Frameworks: Pandas, OpenCV, Agile |
| | • Certifications: SQL, ML with TensorFlow |

EDUCATION

UNIVERSITY OF CALIFORNIA - SAN DIEGO

B.S. Computer Engineering, Minor in Cognitive Science (AI)

Sep/2021-Jun/2025

Relevant Coursework: Advanced Data Structures, Software Tools and Techniques (GitHub, JUnit), Computer Systems (C, Assembly, ARM), Software Team Development (Agile methods, GitHub Projects), Computer Vision (PyTorch, OpenCV), Compiler Construction (ANTLR, MIPS), Components and Design Techniques for Digital Systems (RTL, Netlist, Verilog), Computer Architecture (ALU, Cache, Memory), Advanced Digital Design (ASIC, FPGA), Intro to Embedded Systems (RTOS, SPI, I2C)